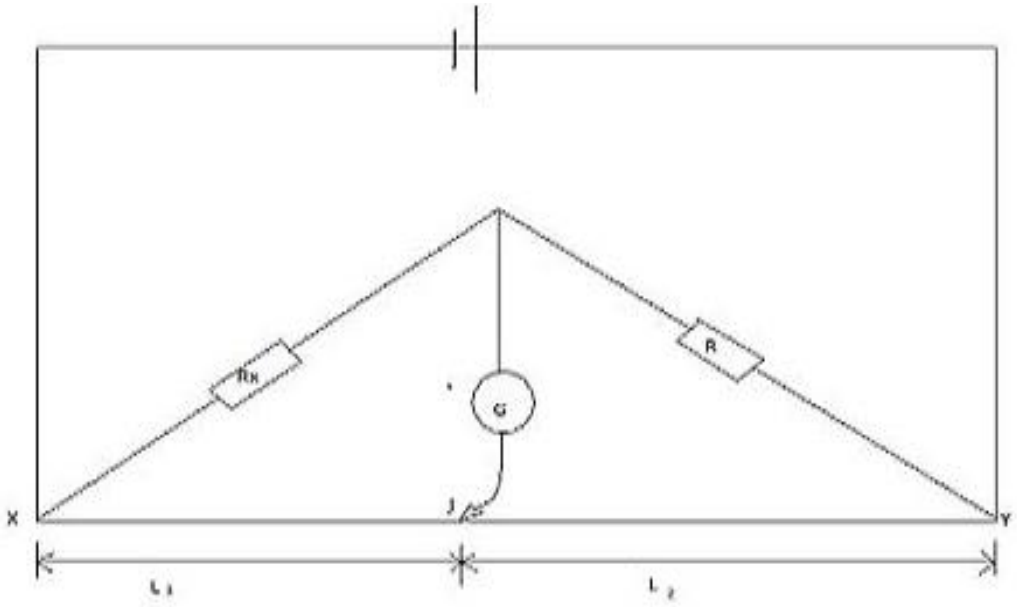
AIM

The aim of the experiment is to use the Wheatstone bridge circuit to determine the resistance of constantan wires of different diameters and the resistivity of a constantan wire.

APPARATUS

Resistance wire board	1
Slide wire measurement bridge	1
Connection box	1
Standard PEK carbon resistor	1
PEK carbon resistor, Q	1
Power supply, 5V/1A, +/-15V	1
Digital multimeter 07134.00	1
Connection cord, 500mm, red	2
Connection cord, 500mm, yellow	2
Connection cord, 500mm, blue	2

DIAGRAM



THEORY

For a Wheatstone bridge circuit, at the balanced point, the potential at C equals the potential at J and no current flows through the digital multimeter. The potential difference across R equals that across the length l_2 and the potential difference across R_x equals that across l_1.

Then, $\frac{R}{R_X} = \frac{l_2}{l_1}$ and $R_x = R\frac{l_1}{l_2}$

The resistance of the material of the wire is proportional to its length and inversely proportional to its cross-sectional area. That is, $R \propto \frac{l}{A}$

$R = \rho\frac{l}{A}$ where ρ is the resistivity of the material of the wire.

PROCEDURE

The experiment was set up as displayed in the manual. The resistances (wire resistances) to be investigated are shown in the diagram above.

Resistor Q and the standard resistor R were connected as shown on the diagram using the connection box. The power supply was switched on and the correction of the connections was checked by touching the bridge wire at the extreme ends X and Y with the slider J. Observing negative and positive readings on the digital multimeter G confirmed the correct connections. The balanced point was located on the wire at J, the position where there is zero reading on the digital multimeter.

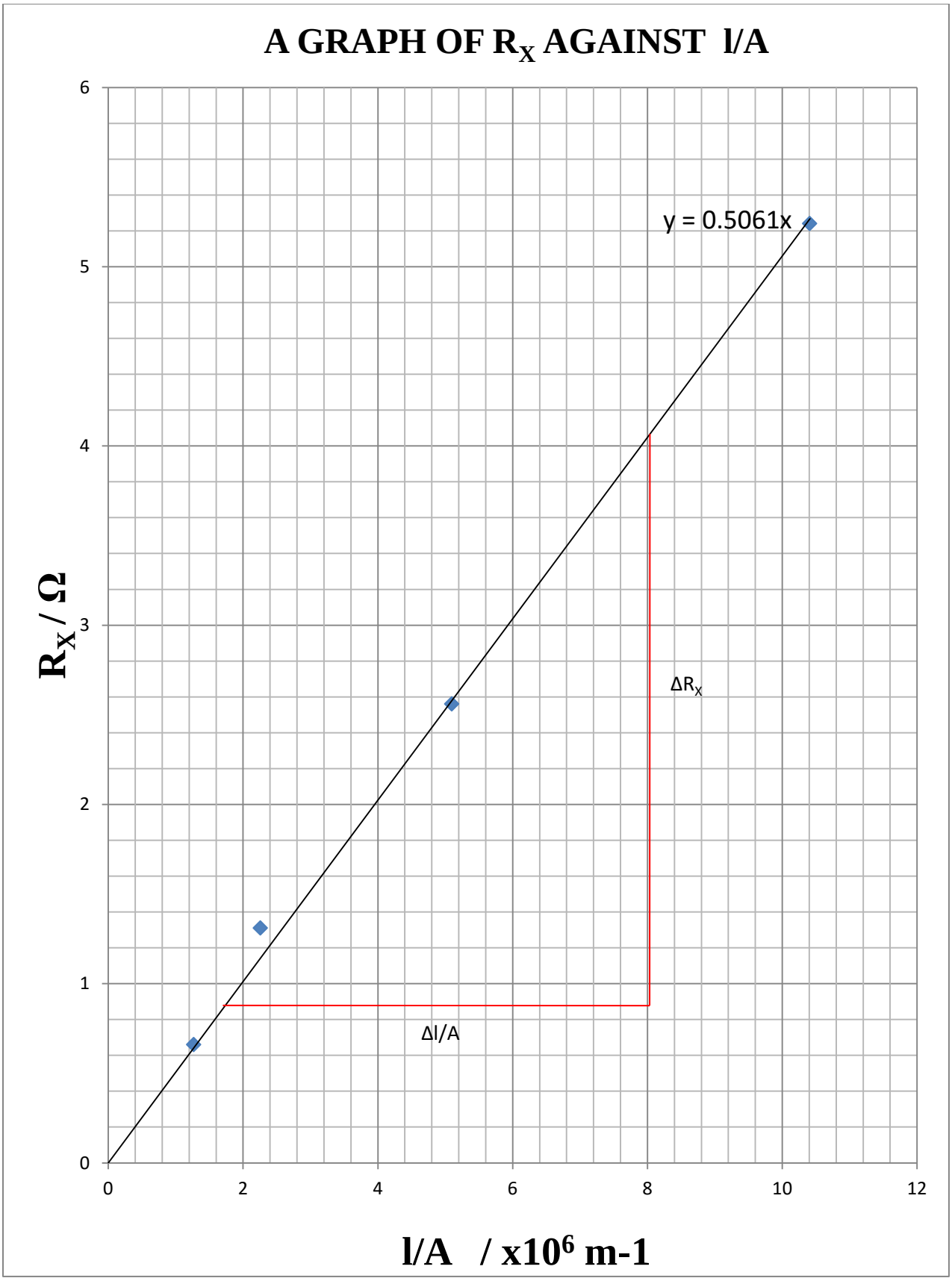
The length l_1 was measured from the end point X to the balanced point and also l_2. The experiment was repeated with the resistance wires of different diameters. The resistances of Q in each of the wires are calculated. Also for each wire, the cross-sectional area was determined. The results were tabulated.

A graph of the resistances of the wires R_x against $\frac{l}{A}$ was plotted where l is the length of the wire.

The slope was determined and its significance indicated.

TABLE OF RESULTS

Diameter of wire	Cross- sectional area / x10^{-7}m	l_1 / mm	l_2 / mm	$R_x = R\frac{l_1}{l_2}$ / Ω	$\frac{l}{A}$ / x10^6 m^{-1}
Constantan 1	7.85	62	938	0.66	1.27
Constantan 0.5	1.96	204	796	2.56	5.10
Constantan 0.75	4.41	116	884	1.31	2.26
Constantan 0.35	0.96	344	656	5.24	10.41


A GRAPH OF R_X AGAINST l/A
y = 0.5061x
ΔR_X
$\Delta l/A$
R_X / Ω
l/A / x10^6 m-1
0
1
2
3
4
5
6
0
2
4
6
8
10
12

CALCULATION

$$slope = \frac{\Delta\Omega}{\Delta\, l/_A} = \frac{4.04\Omega - 0.8\Omega}{8.0 \times 10^6 m^{-1} - 1.6 \times 10^6 m^{-1}} = \frac{3.24\Omega}{6.4 \times 10^6 m^{-1}} = 0.506 \times 10^{-6}\Omega m$$

Comparing $y = mx$ to $R = \rho \frac{l}{A}$ it implies that the slope m is similar the ρ which is the resistivity of the material of the wire.

So, $\rho = 0.506 \times 10^{-6}\Omega m$.

DISCUSSION

Using the diameters of the wires indicated on the resistance wire board, the cross-sectional areas were determined in the formula $\frac{\pi d^2}{4}$ where d is the diameter of the wire. The length of the wire was divided by the cross-sectional areas respectively. Also from the slide wire measuring bridge, the resistance of the wires R_X was also determined by multiplying the standard resistor of 10Ω by the ratio of l_1 and l_2. When a graph of R_X against l/A was plotted, it produced a straight line passing through the origin (0,0). The slope was found to be 0.506 x 10^{-6}Ωm and since the slope is the resistivity in the formula given in the theory above, it means the slope is the resistivity of the material of the wire. In other words, the resistivity of the material of the wire is 0.506 x 10^{-6}Ωm.

SOURCES OF ERROR

1. 1. Most of the time the balanced point fell in a range of about 10mm and so the precise point could have been different from the point recorded.

PRECAUTIONS

1. It was always made sure that the jockey firmly touched wire throughout the reading so as not to obtain a wrong value for the resistance.
2. The connections were always checked after each reading ensure consistent correct value.